

GG migration and cutover to RHEL 7 runbook

- [Pre-requisites](#)
- [pre-checks](#)
- [Cutover steps to RHEL 7](#)
- [Post Cutover](#)
 - [Verify GG sync and ensure no lag between RHEL7 and RHEL6](#)
 - [AWS Steps](#)
- [Rollback](#)
 - [Post Rollback](#)
 - [AWS steps](#)
 -

Pre-requisites

- create new pump on RHEL7 for AWS — [Completed](#)
 - create pump PPSPPP01 to pump trail files from RHEL7 to GG hub (RDS) .
- Switchover RHEL7 to LVDC

pre-checks

- Verify latest veridata compare run to make sure there are NO OOS rows – [Completed](#)
- If its there we need to fix it first before cutover. Below is example : – [NA](#)

```
1. log in to 12c db as GGT user

2. run 15 days veridata compare check OOS rows in veridata for table and OOS rows and columns

3. run below update on target(12c) database as GGT user. Change table name as per OOS report.
update pspadm.PSP_ADDRESS set (ADDRESS_LINE2)=(select ADDRESS_LINE2 from pspadm.PSP_ADDRESS@to_source where
ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1')
where ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1';
```

- Verify GG replication status on both sides

```

1.
GGSCI (pprdpspdb900.ie.intuit.net) 1> info all

Program      Status      Group      Lag at Chkpt  Time Since Chkpt
MANAGER      RUNNING
EXTRACT      RUNNING     EPSPLPD1   00:00:00      00:00:04
EXTRACT      RUNNING     PPSPLPD2   00:00:01      00:00:02
REPLICAT     RUNNING     RPSPLPD2   00:00:00      00:00:07

GGSCI (pprdpspdb901.ie.intuit.net) 1> info all

Program      Status      Group      Lag at Chkpt  Time Since Chkpt
MANAGER      RUNNING
EXTRACT      RUNNING     EPSPLPD2   00:00:04      00:00:06
EXTRACT      RUNNING     PPSPLPD1   00:00:00      00:00:06
REPLICAT     RUNNING     RPSPLPD1   00:00:00      00:00:07

2.
Check lag via heartbeat as well as below:
login to 12c (pprdpspdb901.ie.intuit.net)

ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;

```

Cutover steps to RHEL 7

- stop service and Lock application user accounts on current database (RHEL 6) .

1.

Login to pprdpspdb900.ie.intuit.net and stop prod service

```
SQL> exec dbms_service.stop_service('pspprod');
```

PL/SQL procedure successfully completed.

login to pprdpspdb800.ie.intuit.net and stop DR service

```
SQL> exec dbms_service.stop_service('pspdr');
```

PL/SQL procedure successfully completed.

2.

```
alter user PSPADM account lock;
alter user PSPAPP account lock;
alter user PSPBOADMIN account lock;
alter user PSPFLUXADM account lock;
alter user PSPLOG account lock;
alter user PSPFLUXADMIN08 account lock;
alter user PSPREAD account lock;
alter user PSPTEMP account lock;
alter user APPBUGFIX account lock;
alter user PSPREPORTS account lock;
alter user PSPHIST_QBOE account lock;
alter user ARCADM account lock;
alter user EBIETL account lock;
alter user ICADMIN account lock;
```

Kill existing connections (if still exist)

```
select 'alter system kill session '||''''||sid||','||serial#||''''||' immediate;' from gv$session where
username in ('PSPADM','PSPAPP');
```

3. Verify account_status is LOCKED for all below users

```
select username,account_status from dba_users where username like '%PSP%' or username in
('APPBUGFIX','ARCADM','ICADMIN','EBIETL');
```

USERNAME	ACCOUNT_STATUS
-----	-----
ARCADM	LOCKED
PSPHIST_QBOE	LOCKED
APPBUGFIX	LOCKED
PSPLOG	LOCKED
PSPFLUXADM	LOCKED
PSPFLUXADMIN08	LOCKED
PSPADM	LOCKED
PSPTEMP	LOCKED
PSPBOADMIN	LOCKED
PSPAPP	LOCKED
PSPREPORTS	LOCKED
EBIETL	LOCKED
PSPREAD	LOCKED
ICADMIN	LOCKED

14 rows selected.

- Verify GG replication

```
1.
GGSCI (pprdpspdb900.ie.intuit.net) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPLPD1	00:00:00	00:00:04
EXTRACT	RUNNING	PPSPLPD2	00:00:01	00:00:02
REPLICAT	RUNNING	RPSPLPD2	00:00:00	00:00:07

```
GGSCI (pprdpspdb901.ie.intuit.net) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPLPD2	00:00:04	00:00:06
EXTRACT	RUNNING	PPSPLPD1	00:00:00	00:00:06
REPLICAT	RUNNING	RPSPLPD1	00:00:00	00:00:07

```
2. Check lag via heartbeat as well as below:
login to 12c (pprdpspdb901.ie.intuit.net)

ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.agg_heartbeat;
```

- restart extract EPSPLPD2 on RHEL6
- check and verify sequence between RHEL6 and RHEL7

```
set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;
```

- export and import few db schemas from db901 (RHEL6) to db800 (RHEL 7)

```
1. Take export from db server db901

create or replace directory DATA_PUMP_DIR as '/u01/ogg/exp_bkp';

nohup expdp userid='/' as sysdba' schemas=PSPL0G,PSPTMP,APPBUGFIX,PSPFLUXADM directory=DATA_PUMP_DIR
dumpfile=schemas_%U.dmp log=_schema.log parallel=4 2>& 1 &

2. scp to db server db800

3. Drop the current users on RHEL-7 DB server (db901)
Drop user PSPL0G cascade;
Drop user PSPTMP cascade;
Drop user APPBUGFIX cascade;
Drop user PSPFLUXADM cascade;

4. import
nohup impdp userid='/' as sysdba' schemas=PSPL0G,PSPTMP,APPBUGFIX,PSPFLUXADM directory=DATA_PUMP_DIR
dumpfile=schemas_%U.dmp log=imp_schema.log parallel=4 2>& 1 &
```

- Unlock application user accounts on database (RHEL 7)

```

1.
alter user PSPADM account unlock;
alter user PSPAPP account unlock;
alter user PSPBOADMIN account unlock;
alter user PSPFLUXADM account unlock;
alter user PSPLOG account unlock;
alter user PSPFLUXADMIN08 account unlock;
alter user PSPLOG account unlock;
alter user PSPREAD account unlock;
alter user PSPTMP account unlock;
alter user APPBUGFIX account unlock;
alter user ARCADM account unlock;
alter user EBIETL account unlock;
alter user ICADMIN account unlock;

2. verify
select username,account_status from dba_users where username like '%PSP%' or username in
('APPBUGFIX','ARCADM','ICADMIN','EBIETL');

```

USERNAME	ACCOUNT_STATUS
-----	-----
ARCADM	OPEN
PSPHIST_QBOE	OPEN
APPBUGFIX	OPEN
PSPLOG	OPEN
PSPFLUXADM	OPEN
PSPFLUXADMIN08	OPEN
PSPADM	OPEN
PSPTMP	OPEN
PSPBOADMIN	OPEN
PSPAPP	OPEN
PSPREPORTS	OPEN
EBIETL	OPEN
PSPREAD	OPEN
ICADMIN	OPEN

```

14 rows selected.

```

Enable sequence replication between RHEL 7 and RHEL 6 (one time step)

```

SQL> @sequence.sql
Please enter the name of a schema for the GoldenGate database objects:
ggs
Setting schema name to GGS

UPDATE_SEQUENCE STATUS:

Line/pos
-----
Error
-----
No errors
No errors


GETSEQFLUSH

Line/pos
-----
Error
-----
No errors
No errors


SEQTRACE

Line/pos
-----
Error
-----
No errors
No errors


REPLICATE_SEQUENCE STATUS:

Line/pos
-----
Error
-----
No errors
No errors


STATUS OF SEQUENCE SUPPORT
-----
SUCCESSFUL installation of Oracle Sequence Replication support

```

Post Cutover

Verify GG sync and ensure no lag between RHEL7 and RHEL6

```
send status [extract / replicat]
```

```
GGSCI (pprdqbodb80q.ie.intuit.net as ggs@qboqpd16) 7> info EQBQPD16, detail
```

```
EXTRACT      EQBQPD16  Last Started 2017-11-19 21:24   Status STOPPED
Checkpoint Lag      00:00:02 (updated 00:00:24 ago)
Log Read Checkpoint Oracle Integrated Redo Logs
                  2017-11-28 02:52:22
                  SCN 1976.1787450233 (8488642827129)
```

Target Extract Trails:

Trail Name	Seqno	RBA	Max MB Trail Type
./dirdat/qboqpd16/tr	4692	119910231	500 EXTTRAIL

```
Integrated Extract outbound server first scn: 1976.788177432 (8487643554328)
```

```
Integrated Extract outbound server filtering start scn: 1976.788177432 (8487643554328)
```

Extract Source	Begin	End
Not Available	2017-05-24 20:54	2017-11-28 02:52
Not Available	2017-05-24 20:54	2017-11-19 21:21
Not Available	2017-05-24 20:54	2017-10-26 22:38
Not Available	2017-05-24 20:54	2017-09-26 15:02
Not Available	2017-05-24 20:54	2017-08-03 22:41
Not Available	2017-05-24 20:54	2017-07-13 23:12
Not Available	2017-05-24 20:54	2017-06-09 20:24
Not Available	2017-05-24 20:54	2017-06-09 04:55
Not Available	2017-05-24 20:54	2017-06-08 22:47
Not Available	* Initialized *	2017-05-24 20:54
Not Available	* Initialized *	2017-05-24 20:54

```
Current directory      /u01/ogg/qboqpd16/12.1.2.1.6
```

```
Report file            /u01/ogg/qboqpd16/12.1.2.1.6/dirrpt/EQBQPD16.rpt
Parameter file         /u01/ogg/qboqpd16/12.1.2.1.6/dirprm/eqbqpd16.prm
Checkpoint file        /u01/ogg/qboqpd16/12.1.2.1.6/dirchk/EQBQPD16.cpe
Process file           /u01/ogg/qboqpd16/12.1.2.1.6/dirpcs/EQBQPD16.pce
Error log              /u01/ogg/qboqpd16/12.1.2.1.6/ggserr.log
```

AWS Steps

- **take a snapshot backup** for database in question - psppp001
- stop pump PPITPP01 on RHEL6
- Verify replicat (from RHEL6) status is at EOF on GG hub

```
send status [extract / replicat]
```

- Create new replicat RPSPQPD4 on GG hub for RDS
- stop replicat from RHEL6 and rename parameter file to dummy on GG hub
- start pump PPITPP01 on RHEL6
- Verify OGG monitoring to db800, db901, GG hub server for all GG processes.
- Total proceses on db800 in running status (RHEL7)
 - 1 extract for RHEL6 and AWS
 - 2 pump for RHEL6 and AWS
 - 1 replicat from RHEL6
- Total processes on db901 in running status (RHEL6)
 - 1 extract for monolith

- 2 extracts for IBOB
 - 1 pump for RHEL7
 - 1 pump for AWS
 - 1 replicat for monolith
 - 2 replicats for IBOB
- On AWS in running status
 - 1 replicat for RHEL 7
 - 1 STOPPED replicat for RHEL6
- Test GG replication by insert/update/delete on test table under PSPADM schema

Rollback

- stop application, lock accounts and kill existing session (if required after app shutdown) on RHEL7
- verify GG sync
- restart extract EPSPQPD4 on RHEL 7
- import/export schemas back to RHEL6
- check and verify sequence replication between RHEL6 and RHEL7

```
set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;
```

- Unlock users and start application

Post Rollback

AWS steps

- stop pump PPSPPP01 to AWS on RHEL7
- verify replicat from RHEL7 status is at EOF on GG hub
- stop replicat RPSPQPD4 on GG hub and rename parameter file to dummy
- start old replicat RPSPLPD2 on GG hub
- ~~start old pump PPITPP01 on RHEL6~~
- Test GG replication by insert/update/delete on test table under PSPADM schema